



SeatAI

AI-Powered Classroom Seating Optimization

The seating chart that **plans itself** — in seconds.

SeatAI turns one of the most tedious, judgement-heavy jobs in teaching — arranging a class so every child is set up to succeed — into a one-click decision. It weighs academic balance, behaviour, friendships, diversity and special-needs requirements all at once, and explains every choice in plain language.

Runs 100% in the browser

No accounts · No cloud · No tracking

Works offline

Free & open-source (MIT)

English · العربية · עברית · Русский

THE EVERYDAY PROBLEM

Seating charts are slow, manual, and easy to get wrong



Doing it by hand today

- Teachers spend 30–60+ minutes per class shuffling names on paper or sticky notes.
- Juggling who-can't-sit-with-whom, friendships, eyesight and focus needs is mental overload.
- Special-needs requirements (front row, quiet zone, mobility, EAL/ESL) get missed under pressure.
- Every new term, new student, or behaviour issue means starting over.
- Charts live in one teacher's head — hard to hand to a substitute.



With SeatAI

- A balanced, rule-respecting plan in seconds, not an evening.
- Every constraint is enforced automatically and nothing is forgotten.
- Each placement comes with a plain-language reason teachers can stand behind.
- One click to re-shuffle, rotate, or adjust when things change.
- Print an anonymised chart for a substitute in one tap.

HOW IT WORKS

From class list to optimised chart in four steps

1

Add the class

2

Set the room & rules

3

Optimise

4

Review & print

Type students, import a CSV,
or load a sample class to try it
instantly.

Pick a layout and add any
rules — keep apart, front row,
peer mentors.

A genetic algorithm searches
thousands of arrangements for
the best fit.

Fine-tune by drag-and-drop,
see why each seat was
chosen, export & print.

WHY SCHOOLS ADOPT IT

Six ways SeatAI helps your school



Gives teachers their time back

Replaces an hour of manual shuffling with a few clicks — time that goes back into teaching and planning.



Calmer, better-managed classrooms

Keeps clashing students apart and balances behaviour across the room — fewer disruptions by design.



Inclusion & equity, built in

Front-row, quiet-area, mobility and EAL/ESL needs are first-class rules the optimiser always honours.



Student data never leaves the device

No servers, no accounts, no analytics. A clean answer to data-protection questions.



Zero cost, zero IT burden

Free and open-source static app with no backend to run — and it works fully offline.



Consistent & defensible decisions

Every placement has a recorded rationale — for parents, colleagues, or leadership.



Privacy & IT: the easy approval

SeatAI is designed so there is simply nothing to leak and nothing to provision.

- ✓ **Local-only data.** Rosters, notes & photos stay in the browser (IndexedDB).
- ✓ **No tracking.** No analytics, telemetry, or third-party scripts.
- ✓ **Optional AI is opt-in.** Keys stay local; calls go browser-direct.
- ✓ **No accounts.** No login, email, or sign-up — open the URL and work.
- ✓ **No backend.** Pure static HTML/CSS/JS — host it on anything.
- ✓ **Anonymised printing.** One toggle prints initials-only for subs & visitors.

WHO IT'S FOR

Value at every level



Teachers

- End the seating-chart chore



School leaders

- Supports inclusion & behaviour goals



IT & data leads

- Nothing to host or secure

- Confident, explainable placements
- One-click rotation each term
- Drag-and-drop fine-tuning

- Consistency across classrooms
- No licence cost to budget for
- Defensible, recorded decisions

- No student data off-device
- Deploy & forget — works offline
- Modern-browser support floor

EVERYTHING INCLUDED

A complete toolkit — no add-ons, no tiers

- ✓ Rows, clusters, U-shape, circle & custom layouts
- ✓ Multi-start genetic optimiser (Fast / Balanced / Best)
- ✓ Drag-and-drop, click, or keyboard seat swaps
- ✓ Keep-together / keep-apart & peer-mentor rules
- ✓ Front-row, back-row, aisle & window constraints
- ✓ Special-needs & EAL/ESL aware placement
- ✓ Per-student photos & teacher notes
- ✓ Plain-language & optional AI placement reasons
- ✓ CSV import/export · PDF · image · print
- ✓ Multi-class projects with backup & restore
- ✓ Undo / redo on every change
- ✓ Seating rotation tracker
- ✓ Visual diff & A/B comparison of plans
- ✓ Heat-maps: gender, conflicts, scores
- ✓ 2D & 3D classroom views
- ✓ Light / dark theme & adjustable text size
- ✓ Installable offline PWA
- ✓ Full keyboard & screen-reader accessibility
- ✓ English, Hebrew, Arabic & Russian (RTL)
- ✓ Works on desktop, tablet & phone

Try it with your own class in 2 minutes

Open SeatAI, load a sample class or import your roster, and click Optimise.
No install, no sign-up, and no data leaves the room.

Free & open-source under the MIT licence · Deploy school-wide at no cost